A Failed Reconstruction of the SLRTP2025 Back-Translation Evaluator from Public Artifacts

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Abstract

We audit whether the released SLRTP2025 back-translation (BT) evaluator for sign language production can be reconstructed from its public artifacts, an evaluation-only repository without training scripts. Our reconstruction attempt failed.

Twenty-eight recipe-conditioned runs—fourteen legacy-implementation seeds and fourteen near-faithful runs (four validation-freq-misread, ten step-corrected)—reach dev BLEU 6.5–10.5 versus 13.38, teacher-forced token accuracy 55–60% versus 96.5%, and training-pool readout 7.2–9.3 versus 78.8. Because all constructible checkpoints are less competent, we cannot distinguish whether the non-reproduction of a constructed adversarial replay-probe response (a post-selected descriptive estimate of **+10.24** sacreBLEU on the released evaluator, selected as the maximal-gap cell among eight explored probes) reflects the competence gap, an unrecovered training mechanism, or their interaction.

Reference replacement with human back-translations is associated with a gap reduction to **+0.95** (from **+10.03**) on a 461-item subset—a single-annotator point estimate that cannot isolate reference-text defects from lexical-coupling disruption, paraphrase shift, or BLEU floor effects. Donor-pool substitution reverses the sign; a pool-size-controlled resampling (1,000 draws of 640-vs-640 donor pools) gives a residual pool-origin contrast of **+8.38** BLEU (95% interval [**+7.44, +9.34**]), an association rather than an identified training-exposure effect.

These findings constitute a failed reconstruction from the public artifacts under our disclosed implementation, not a causal checkpoint effect, a general leaderboard-instability claim, or a proof that the public materials themselves are insufficient (the latter would require an independently validated reference implementation, which we could not obtain). We frame the probe as an adversarial lexical-coupling stress test throughout.

Keywords: sign language production, back-translation evaluation, reproducibility audit, reference-frame sensitivity, learned evaluation metrics, PHOENIX-2014T

1 Introduction

Sign language production (SLP) systems are commonly evaluated by back-translation (BT): a trained sign recognizer decodes each output pose sequence, and a text metric compares the decoding with the reference. This measures recognizer-readable content through a learned measurement model, so system rankings may reflect both the evaluated outputs and the inductive biases of a particular evaluator checkpoint. The present audit is framed as a *public-artifact reconstruction attempt*: whether our implementation, using the released SLRTP2025 public artifacts, can reconstruct the released BT evaluator. We address three research questions:

RQ1. Can the public artifacts reproduce the released evaluator's competence and training-pool readout?

RQ2. (Exploratory, competence-unmatched.) Does a frozen adversarial replay probe's ordering reproduce across evaluator checkpoints constructed from the public recipe? This is a *descriptive secondary observation*: all constructible checkpoints are less competent than the released evaluator, so any non-reproduction cannot be attributed to checkpoint identity versus competence versus an unrecovered training mechanism.

RQ3. How sensitive is the probe response to the reference text and to donor origin?

We do *not* frame the work as a test of checkpoint identity. Every checkpoint we can construct from the public recipe is less competent than the released evaluator (14 paper-derived reconstructions: dev BLEU-4 6.5–9.4; 14 near-faithful runs: 7.6–10.5; broader constructible family training-log max 10.5; all vs. released 13.38), so competence and checkpoint identity are confounded. We therefore frame RQ2 strictly as a *descriptive non-reproduction*: among independently constructed but uniformly less-competent checkpoints, the ordering does not reproduce. Without a competence-matched independent implementation—which we could not obtain—we cannot distinguish whether this reflects the competence gap, an unrecovered training mechanism, or their interaction. The replay probe itself is an *adversarial lexical-coupling stress test*: the source-conditioned retriever selects donors by surface vocabulary overlap with the German source, which is identical to the scoring reference in PHX-public, so the probe deliberately targets the strongest form of the shortcut rather than a deployable retrieval pipeline.

Prior work shows BT scores can be modest for recorded signing and can improve without faithful target conditioning Walsh et al. (2024); Hong and Košecká, Jana (2026) (Hong and Košecká, arXiv preprint, preliminary); sign-native or pose-aware metrics provide complementary evidence Kim et al. (2024); Imai et al. (2025); Jiang et al. (2025), yet BT remains central to SLRTP2025 Walsh et al. (2025). Chen et al. Chen et al. (2022) showed that learned BERT-based evaluation metrics can fail to reproduce due to undocumented preprocessing, missing code, and baseline implementation differences—a methodology directly relevant to our finding that our public-artifact reconstruction attempt failed to reproduce the released checkpoint. An unresolved question is whether a retrieval–reference stress-test ordering persists across independently trained instances of the same BT architecture.

This question is exposed by an adversarial retrieval–reference probe on PHX-public: under the released evaluator, source-conditioned whole-donor retrieval scores 23.02 sacreBLEU (0–100 scale) versus 12.78 for the recorded test poses. Because the retriever selects donors by lexical overlap with the source sentence—which is identical to the scoring reference in PHX-public—this probe is a deliberately constructed adversarial stress test, not evidence of general evaluator preference or better signing.

We address the three research questions with the following evidence (full results in §4.2–§4.3.1). For RQ1, none of our reconstructions reached the released evaluator's competence. For RQ2, all 61 non-degenerate constructible checkpoints give negative probe gaps (range $[-2.01, -0.21]$); we cannot determine whether this reflects the competence gap or a checkpoint-identity effect. For RQ3, reference replacement with human back-translations is associated with a gap reduction; donor-pool substitution reverses the sign. The source-conditioned retriever shares surface vocabulary with the original reference (§3.2), so the RQ3 result conflates reference-text properties, lexical-coupling disruption, paraphrase shift, and BLEU floor effects.

Why a constructed probe matters.

Retrieval-augmented SLP entries draw donors from the evaluator's training pool, so the adversarial probe isolates a risk that production systems face in diluted form. The audit contributes a transferable template (provenance table, checkpoint registry with SHA-256 deduplication, evaluator-replication controls, teacher-forced-vs-free-decode diagnostics) for public-artifact reproducibility testing. The specific probe response is a PHX-public released-evaluator case study and does not support general claims about BT evaluation or retrieval-based SLP.

2 Related Work

Note on preprints. Where we cite 2025–2026 arXiv preprints Hong and Košecká, Jana (2026); Cory et al. (2026); Alkain et al. (2026); Czehmann et al. (2026); Artiaga et al. (2025); Desai et al. (2024), we treat their findings as preliminary reports rather than peer-reviewed results; conclusions in this audit do not hinge on the correctness of any single preprint. All arXiv citations were accessed on 2026-08-10.

Learned evaluation in sign language production.

Neural SLP often evaluates with a frozen sign recognizer Camgoz et al. (2018); Saunders et al. (2020); BT was proposed as an SLP evaluation protocol by Saunders et al. Saunders et al. (2021), who argued it shifts absolute scores but not relative rankings between similar configurations — a proposition our audit does not directly test (its similarity condition is unmet in our family) but re-examines for a released evaluator on which a retrieval-probe reversal appears. SignBLEU Kim et al. (2024), SiLVER-Score Imai et al. (2025), and pose-native measures offer complementary views, but

BT BLEU remains central to SLRTP2025 Walsh et al. (2025). Although newer metrics provide better human-correlated evaluation Jiang et al. (2025); Cory et al. (2026) (BackTranslation2.0, arXiv preprint), the SLRTP2025 BT evaluator is the *standardized infrastructure* on which the challenge ranking depends; auditing its reproducibility and failure modes is infrastructure maintenance, not a claim that decoded BT-BLEU is the best available metric.

Retrieval and shortcut risks.

Retrieval-based SLP ranges from phonetic composition to retrieval-enhanced generation Walsh et al. (2024,?); Zuo et al. (2025); Tasyurek et al. (2025), and SLRTP2025's leading entry is retrieval-based, making donor copying pertinent Walsh et al. (2025). Memorization results in retrieval-augmented language modeling show strong task scores can reflect reproduction of training examples Carlini et al. (2021); Kandpal et al. (2022); Khandelwal et al. (2020); a shortcut under one learned evaluator does not show the ordering persists under another checkpoint.

Robustness of learned metrics and corpus audits.

Hong and Košecká Hong and Košecká, Jana (2026) *report* (preprint) that BT-BLEU can diverge from target faithfulness; Walsh et al. Walsh et al. (2024) document a low recorded-pose BT baseline; Jiang et al. Jiang et al. (2025) recommend BT likelihood over decoded BLEU; and model-based generation metrics are further criticized by Yazdani et al. (2026); Cory et al. (2026); He et al. (2024). Two 2026 PHOENIX-2014T audits bear directly on our reversal: Czehmann et al. Czehmann et al. (2026) released human back-translations and documented reference information loss, Alkain et al. Alkain et al. (2026) showed the highest train–test text overlap among tested SLT corpora, and Artiaga et al. Artiaga et al. (2025) showed signer-dependent evaluation overestimates SLT capability. None runs multi-checkpoint evaluator experiments or donor-pool substitution controls.

Reproducibility of learned evaluation metrics.

Chen et al. Chen et al. (2022) showed that reproducing BERT-based evaluation metrics can fail due to undocumented preprocessing, missing code, and baseline implementation differences, even when model weights are released—a finding methodologically close to our public-recipe non-reproduction result. The difference is that our audit targets a full BT evaluator pipeline (architecture, training recipe, and checkpoint) rather than a metric alone, and the non-reproduction is demonstrated across 61 unique non-degenerate decoded runs (all 61 with strictly negative gaps; five degenerate excluded), corresponding to 60 distinct SHA-256 weight binaries after one validation-freq-misread/long-schedule seed-202 collision is deduplicated.

PHOENIX-2014T as a language resource: known limitations.

PHOENIX-2014T Camgoz et al. (2018) (8,257 weather-forecast interpretations by 9 signers) is the most widely used DGS parallel corpus, but four properties constrain our audit's generalizability: (i) template density inflates train–test n-gram overlap Alkain et al. (2026) (32 items whose reference exactly matches a training text account for a disproportionate share of the gap); (ii) reference texts are speech transcriptions, not DGS translations Czehmann et al. (2026); (iii) the 178-keypoint pose materialization loses fine handshape appearance and non-manual detail (21 landmarks per hand plus 128 face points capture joint centres but not finger shape, palm orientation, mouth shape, or eye gaze; our MSKA control uses 133 keypoints; SI Sup. L); (iv) signer-dependent evaluation can overestimate capability Artiaga et al. (2025). The lexical coupling between source and reference amplifies these effects in our adversarial probe. The transferable audit components (provenance table, checkpoint registry, leakage controls) apply to any learned evaluator paired with a public training corpus.

Table 1: Novelty relative to closely related PHOENIX-2014T and learned-metric reproducibility studies.

Study	Primary finding	Method	Distinction from present work
Czehmann et al. (2026)	Reference replacement changes BLEU scores and rankings on PHOENIX-2014T	Human back-translations of test set	We apply the matched-subset protocol to the released evaluator with a frozen replay probe and quantify paired attenuation
Alkain et al. (2026)	Highest train–test text overlap among tested SLT corpora	N-gram overlap quantification	We use their finding to contextualize why PHX-public is particularly susceptible to reference coupling
Artiaga et al. (2025)	Signer-dependent evaluation overestimates SLT capability	Signer-disjoint splits	We include signer as a regression covariate (not signer-disjoint evaluation; ΔR^2 from removing the signer block is $\leq 1\%$)
Jiang et al. (2025)	BT likelihood is a more stable correlate of human judgment than decoded BLEU	Meta-evaluation of pose/embedding/BT metrics	We report that teacher-forced likelihood does not reproduce the reversal (REC NLL 3.060 vs PURE 3.479)
Chen et al. (2022)	BERT-based metrics fail to reproduce due to missing preprocessing/code	Multi-implementation replication	We extend the learned-metric reproducibility question to a full SLP evaluator pipeline with 68 training runs across 11 families
This work	Our disclosed reconstruction from the public artifacts did not attain the released evaluator's competence or replay-probe behavior	68 training runs across 11 families, fixed-output cross-evaluator re-decoding, reference/donor substitution	—

3 Methods

3.1 Evaluation design

Let E_c denote a BT evaluator checkpoint, x_n the source sentence (German) for item n, and $o_{i,n}$ the pose sequence for output condition i and item n; E_c produces $h_{c,i,n} = E_c(o_{i,n})$ and $B_{c,i} = M(\{(h_{c,i,n}, y_n)\}_{n=1}^{N})$, where y_n is the reference text and M is the corpus-level sacreBLEU functional unless stated otherwise. We call a result a *descriptive non-reproduction* when a probe response observed under the released evaluator does not appear across independently constructed checkpoints under a fixed output set; this term is descriptive and does not imply a causal checkpoint-identity effect (which we cannot test because all constructible checkpoints are less competent). The fixed primary set comprises recorded test poses (REC), random donor copy, source-conditioned whole-donor retrieval, a Progressive Transformer generator, and retrieval–generator compositions; oracle-gloss variants are diagnostic controls.

Operational definition of reconstruction competence.

We define an *author-operational* competence gate—not a public-sufficiency judgment—as dev BLEU-4 ≥ 12.38 *and* dev WER ≤ 86.37 (one-sided tolerance of ± 1 BLEU / ± 3 WER around the released evaluator's 13.38 / 83.37). This gate is a sensitivity summary, not a community standard: the gate conclusion is insensitive across a $[0.5, 2.0]_{\text{BLEU}} \times [1.5, 4.5]_{\text{WER}}$ grid (§4.3), but other tolerance choices would give other pass/fail counts. The primary evidence for the reconstruction failure is the *continuous* metric gap itself (Table 4: dev BLEU-4 6.5–10.5 vs. 13.38; train readout 7.2–9.3 vs. 78.8; teacher-forced token accuracy 55–60% vs. 96.5%); the gate merely summarises this continuous evidence into a binary pass/fail for table presentation. The source of the gap may reflect information incompleteness in the public artifacts, stochastic underspecification, or a defect in our disclosed implementation; without the original training command, these cannot be separated.

The main evaluation uses PHX-public, a German weather-domain text-to-pose benchmark whose released SLRTP2025 pose records contain 7,060/515/641 sequences; one official test ID is absent from the released materialization, so every result is conditioned on the complete 641-record release (SI Sup. A). Table 2 gives the three-way provenance: the official PHOENIX-2014T split, the released pose materialization, and the released evaluator's actual training manifest (confirmed via `config.yaml`, whose manifest matches the shipped `.pt` tensors). Supplementary analyses use **PHX-holdout-MSKA** (133-joint, 1,538 windows) and CSL-Daily (Mode B, 1,176 sequences; SI Sup. L); the protocol summary is in SI Table S28.

All pose sequences are evaluated at 12.5 fps, matching the SLRTP2025 evaluator training distribution. The data flow is: official PHOENIX-2014T video at 25 fps $\rightarrow$ SLRTP2025 skeleton extraction at 25 fps $\rightarrow$ our `[::2]` subsampling to 12.5 fps for evaluator input; the official evaluation command uses `--fps 25` (the input file's native rate), but the evaluator was trained on 12.5 fps-subsampled poses, so our subsampling matches the training distribution rather than the command-line flag. A three-path regression confirms no double subsampling (details in SI Sup. F). PHX-public uses α=0.88 (selected on dev); the supplementary splits use α=0.5.

Table 2: Three-way data provenance for PHX-public. The released evaluator's training manifest matches the released pose materialization (7,060/515/641), not the full official PHOENIX-2014T split. All 41 missing IDs carry the missingness label `absent_from_slrtp2025_released_pose_records`—a status flag rather than an explained mechanism (SI Sup. A). SHA-256 hashes verify file integrity.

Split source	Train	Dev	Test
Official PHOENIX-2014T	7,096	519	642
Released SLRTP2025 pose (`.pt`)	7,060	515	641
Released evaluator `config.yaml` manifest	7,060	515	641
Missing from official	36	4	1

3.2 Systems and retrieval controls

For each target, source-conditioned whole-donor retrieval scans the complete training pool after Unicode NFKC normalization, whitespace normalization and lowercasing, selecting the sequence with maximum source-text token-set Jaccard similarity. Oracle-gloss retrieval instead uses target gloss and is treated as an input-leakage diagnostic; random retrieval selects a donor uniformly. Donor exclusion removes exact normalized text or gloss duplicates or donors above threshold τ; gloss-based exclusion is therefore an offline diagnostic, not a deployable inference procedure.

Single canonical materialization. Every number in this paper is generated from one canonical donor registry built by the deterministic §3.2 algorithm (exact-normalized-text exclusion applied), whose SHA-256 is `9170a53026ab3263b451a3632a9e02-318745acabf12f0b679921477afbefa301`. Reproduction is supported at four layers (§ Sup. Q, Table S36): (i) **L1—recompute selected reported statistics**: a single command (`make core-audit`) regenerates the registry and verifies headline numbers from committed per-item decodes; it does *not* retrain models or decode from checkpoints, and does not verify every number in the paper; (ii) **L2—analysis re-execution**: each analysis script re-executes from the committed per-checkpoint data and hypotheses; (iii) **L3—model re-evaluation**: re-decoding fixed outputs from checkpoint weights requires the training bundle; (iv) **L4—from-scratch retraining**: full retraining requires the restricted PHOENIX-2014T licence, the exact environment, and approximately 200 GPU-hours. The `make core-audit` command executes L1 only and is not an end-to-end reproduction. A claim-to-file manifest mapping each headline number to its source file and verification command is in `claim_manifest.json` (artifact root). An earlier pre-exclusion materialization (PURE 23.79, gap +11.01) survives only as a sensitivity analysis in SI Sup. N.

Lexical coupling of the source-conditioned probe (adversarial design).

In PHX-public, the source sentence x_n (German weather report) is identical to the reference text y_n^{orig} at the sequence level (the reference is a speech transcription of the same German sentence). The TN-PURE-v1 retriever selects training-pool donors by token-set Jaccard similarity between x_n and training texts, so it preferentially retrieves donors whose accompanying training texts share surface-level vocabulary with both the source *and* the reference. Scoring a donor copy against this lexically coupled reference under a learned evaluator therefore confounds donor similarity, evaluator readout, and reference overlap. We explicitly frame this probe as an *adversarial stress test* targeting the strongest form of the lexical-coupling shortcut—not as evidence of general evaluator preference, deployable retrieval quality, or better signing. This framing has two consequences: (i) the +10.24 gap is the expected output of a deliberately adversarial design, not a discovery about the evaluator's ranking behavior; (ii) the human-reference attenuation analysis (§4.3.1) reflects disruption of the lexical coupling as well as reference-text information loss, paraphrase shift, and BLEU floor effects—not a clean separation of any single mechanism.

Retrieval composition.

TN-PURE-v1 uses whole-donor copy with [::2] subsampling to 12.5 fps, preserving the donor's original duration. TN-PTCOMP-v1 (a secondary control) composes a PT scaffold with donor hands and face via α-blending of upper-body joints (α=0.88; full joint-set definitions in the artifact). The matched seen–unseen factorial retains 605/641 targets matched on duration and word count.

Donor-origin contrast retrieval systems.

To test whether the probe response co-occurs with donor origin, we construct frozen probe systems: **UNSEEN-PURE-v1** retrieves from the 640 other test poses; **CROSSFIT-PURE-A/B** split queries by hash into two folds; **SEEN-PURE-MATCHED-v1** selects a train-pool donor within ±0.1 Jaccard of each UNSEEN donor (622/641 admit one). SEEN-PURE-v1 re-uses the canonical TN-PURE-v1 registry; these are constructed probes, not deployable generators (construction rules in the artifact).

3.3 Evaluator checkpoint family

The main inferential checkpoint family comprises the original frozen SLRTP2025 evaluator and 14 *recipe-conditioned reconstruction* runs trained from scratch with the documented architecture on a common frozen partition (six prospectively specified legacy seeds 101–606 plus eight post-hoc extension seeds 707–1405, all under the identical paper-derived protocol). The original bundle's `validations.txt` logs 202 validation points over 2,828 optimizer steps (≈102 epochs), with decoded-BLEU selection reaching best dev BLEU-4 13.38 at step 1,820; the trajectory is analyzed in §4.3.1, and intermediate checkpoints are not published. We use two reconstruction protocols — *paper-derived* (14 runs: 6 prospectively specified seeds 101–606, 8 extension seeds 707–1405; legacy script, 300 epochs, NLL selection) and *near-faithful* (14 runs: 4 validation-freq-misread + 10 step-corrected; decoded-BLEU selection on step-level validation every 14 optimiser steps, matching the released `validation_freq=14` JoeyNMT semantics) — sharing the documented architecture (3L/256/512ff, 8 heads), tokenizer, manifests (7,060/515), [:: 2] subsampling, Adam (1e-3, wd .001), batch size 256, and plateau ×0.8 LR schedule (full provenance in SI). In total 68 from-scratch training runs across 11 families (registry in SI Sup. D); 66 were beam-3 decoded on PHX-public under the canonical §3.2 donor registry, corresponding to 65 unique weight binaries — one validation-freq-misread seed-202 artifact is byte-identical to a long-schedule artifact (same wandb run, identical per-step losses and best-checkpoint SHA-256). Of the 66 decoded training runs, five are degenerate (three α=1.0 distillation students with empty hypotheses, two smallest ladder fractions with BLEU ≈ 0) and are excluded, so the non-replication result rests on 61 non-degenerate decoded training runs, all 61 with strictly negative gaps. Table 8 gives the full accounting.

What the 68 runs are—and are not.

The 68 runs are *not* 68 same-recipe replication attempts. Table 3 maps each family to the public source it draws on, the known deviation from the original training, the number of runs, the checkpoint-selection rule, and the resulting dev BLEU-4 range. The 10 step-corrected runs and the 4 validation-freq-misread runs approach a faithful reconstruction of the released recipe; the remaining 54 runs are diagnostic, sensitivity, or competence-extrapolation analyses (joint-loss greedy/beam-3, distillation, ladder, rescue, confirmation, long-schedule). The public-recipe non-reproduction conclusion rests on the 14 near-faithful runs (of which 10 are fully step-corrected), while the remaining families serve as triangulating evidence that no alternative construction we tested reproduces the reversal.

Training-pool readout (descriptive observation).

We define *training-pool readout* as the in-sample autoregressive free-decode BLEU and exact-match (EM) rate on the full 7,060-item training set under beam-3 decoding—not teacher-forced accuracy. The released evaluator decodes its full 7,060-item training pool at free-decode BLEU 78.8 with 70.7% exact-match, while retaining dev/test performance of 13.38/12.78. The constructible family's readout is far lower (reconstructions: 7.2–9.3 BLEU, 1.5–2.3% EM; distillation: 6.6–8.9 BLEU, 1.3–1.9% EM); no constructible checkpoint approaches both simultaneously. The training-pool free-decode BLEU ratio (released: $78.8 / 13.38 \approx 5.9$ vs. reconstruction family: 0.93–1.12) quantifies how much the model reads out above its in-domain performance. We report these as continuous descriptive observations: the released evaluator was observed first, the family range afterwards, and any threshold is post-hoc.

3.4 Reconstruction sanity checks

Because the 28 recipe-conditioned runs (14 paper-derived + 14 near-faithful) share the same codebase, data pipeline, and evaluation protocol, we performed three checks on them to rule out several basic plumbing failures (data loading, masking, decoding alignment, vocabulary mismatch, pose-subsample direction)—a strictly weaker claim than ruling out a systematic implementation defect (the 47 diagnostic runs—distillation, rescue, ladder, long-schedule, large-arch—and 3 holdout controls deliberately vary the objective, split, architecture, or hyperparameters, and are not claimed to share it):

ll-sample overfit test. A reconstruction model trained on 50 and 100 sequences overfits to free-decode BLEU 100.00 (NLL $\to$ 0.009), ruling out defects in loading, masking, or decoding (`results/overfit_test.json`).

Table 3: Public-artifact provenance of each reconstruction family. "Dev BLEU-4" is the uniform full-pool beam-3 value (§3.4); the released evaluator scores 13.38. "Selection" is the checkpoint-selection rule. The 14 paper-derived runs use NLL selection on a legacy script; the step-corrected runs match the released `validation_freq=14` JoeyNMT semantics (validation every 14 optimiser steps) with decoded-BLEU checkpoint selection; the joint-loss variants additionally correct the recognition-loss omission. Only recipe-faithful families support the public-recipe non-reproduction claim; the rest are diagnostic.

Family	Public source used	Known deviation	Runs	Selection	Dev BLEU-4
Paper-derived (prosp.-spec. + ext.)	config.yaml + slt architecture	legacy script; 300 epochs; NLL selection	14	NLL on dev	6.5–9.4
Validation-freq-misread	config.yaml (decoded-BLEU)	`validation_freq=14` misread as epochs initially	4	decoded BLEU	8.3–9.8
Step-corrected (re-trained)	config.yaml + JoeyNMT step semantics	fixed `validation_freq` interpretation (14 steps) + decoded-BLEU selection	10	decoded BLEU	7.6–10.5
Joint-loss (greedy)	config.yaml (`recognition_loss_weight=1.0`)	adds CTC recognition loss to translation CE	8	greedy-BLEU	6.3–10.2
Joint-loss (beam-3)	config.yaml (eval_beam=3)	same as greedy but beam-3 selection	8	beam-3-BLEU	8.7–10.5
Confirmation	recipe-faithful protocol	independent re-run for reproducibility	2	decoded BLEU	9.0–9.5
Long-schedule	config.yaml (epochs=3000)	extended training budget	1	decoded BLEU	≤10.02
Rescue (dropout/wd)	hyperparameter grid	dropout/wd variations; none passed gate	1	decoded BLEU	≤10.02
Distillation	released checkpoint as teacher	student–teacher KL; inherits released weights	15	decoded BLEU	7.0–9.0
Ladder (data fractions)	config.yaml	data-subset fractions (0.125–0.75×)	4	decoded BLEU	≤9.8
Joint-loss (supplementary)	exploratory v2 seeds (1809/1810)	out-of-protocol v2 explorations	1	—	not decoded
Total			**68**		

nified inference regression. Released and reconstruction checkpoints load through the identical pipeline with byte-identical vocab files (SHA-256 confirmed); the released checkpoint reproduces canonical REC 12.78 exactly.

her-forced vs. free-decode. Table 4 shows the released evaluator achieves 96.5% teacher-forced train token accuracy and 78.8 free-decode train BLEU, while reconstructions achieve 55–60% and 7.2–9.3. The gap is present in both teacher-forced and free-decode metrics on both train and dev sets.

These three checks do *not* rule out subtler defects that the overfit and inference tests cannot probe: training objective composition (e.g., the recognition/CTC term that our v1 protocol omitted despite `recognition_loss_weight=1.0`), loss normalization, CTC target handling, gradient accumulation semantics, scheduler timing relative to optimizer steps, checkpoint-selection argmax vs. lexico-order tie-breaking, dropout schedule, label smoothing, weight initialisation, or AdamW vs. Adam semantics. The four protocol mismatches documented in §3.4 already demonstrate that our v1 training implementation was not initially faithful to the released config; we therefore frame RQ1 below strictly as "our disclosed reconstruction did not attain the released competence", not as "the public materials themselves are insufficient". The two would be equivalent only if a reference implementation independently reproduced (or failed to reproduce) the released checkpoint from the same public artifacts—an independent verification we could not obtain (the released repo is eval-only; SI Sup. R).

Table 4: Teacher-forced and free-decode diagnostic statistics for the released evaluator and the fourteen reconstruction seeds (full table in `results/checkpoint_stats.json`). Teacher-forced values are on 500 randomly sampled items (seed 42, sampled without replacement; the sampling SD across 5 re-samples is $\leq$0.3 pp for token accuracy and $\leq$0.01 for NLL/tok); free-decode values are full-pool (7,060 train / 515 dev) under the uniform beam-3 protocol (§3.4). sacreBLEU `signature: BLEU|nrefs:1|case:mixed|eff:no|tok:13a|smooth:exp|version:2.5.1` (max n-gram order 4, the default).

Checkpoint	Train tok. acc.	Train NLL/tok	Train free-decode BLEU	Dev NLL/tok	Dev free-decode BLEU
Released	0.965	0.158	78.82	3.236	13.38
Reconstructions (range)	0.550–0.596	1.72–1.92	7.20–9.25	2.63–2.68	6.49–9.37
Reconstructions (mean)	0.572	1.82	8.35	2.66	8.13

Training-implementation diagnostics.

The reconstruction training logs (`checkpoints/*/training_log.json`, n=67) record per-epoch train/dev NLL and learning rate. All recipe-conditioned runs use effective batch size 256 (physical batch 64 $\times$ gradient accumulation 4), Adam (1e-3, weight decay 1e-3), and the plateau $\times$0.8 schedule on dev NLL, with identical config (`config_sha256 0321104a...`) and byte-identical vocabularies to the released evaluator. The median train-NLL trajectory is healthy (epoch 1: 5.83; epoch 5: 2.99; epoch 10: 2.54; epoch 25: 2.09 across 28 reconstruction logs), confirming normal optimization. Per-step loss, gradient norm, and per-epoch teacher-forced token accuracy are not archived (only per-epoch NLL aggregates were logged); we therefore cannot rule out a subtle scheduler-timing or normalization difference, which is consistent with our scoped "failed reconstruction attempt" framing. A 25$\rightarrow$12.5 fps pose-subsampling regression test (`scripts/test_fps_regression.py`) passes on 8 released pose tensors, ruling out double subsampling. Full diagnostics in `results/robustness_diagnostics.json` and `results/fps_regression_test.json`.

Protocol robustness: four training-protocol variants.

The original reconstruction training (`train_matched.py`) differs from the released `config.yaml` in four respects: (i) the loss omits the recognition/CTC term despite `recognition_loss_weight=1.0`; (ii) dev validation runs once per epoch rather than every `validation_freq=14` optimizer steps (JoeyNMT semantics); (iii) the `--selection bleu` branch's `improved` flag is hard-coded `False`, so it never saves a BLEU-best checkpoint; (iv) the released checkpoint uses beam-search evaluation while our original runs used NLL or greedy decode. To isolate whether these differences explain the competence gap or the non-reproduction, we trained four protocol variants on the same released config, architecture, data, and vocabularies:

- *Translation-only / epoch-val / NLL* (original; seeds 101–606, 707–1405, plus validation-freq-misread 101–404): translation cross-entropy only; epoch-level validation; NLL checkpoint selection. This is the protocol used for the 14 paper-derived and the 4 validation-freq-misread runs.
- *Translation-only / step-val / decoded-BLEU* (seeds 1701–1708, 505, 606): corrects (ii)–(iii)—still translation-only loss, but step-level validation every 14 optimiser steps and decoded-BLEU checkpoint selection via greedy decode. These are the 10 "step-corrected" near-faithful runs.
- *Joint-loss / step-val / greedy-BLEU* (seeds 1801–1808): corrects (i)–(iii)—joint CTC recognition loss (w=1.0) plus translation CE; step-level validation every 14 optimizer steps; decoded-BLEU checkpoint selection via greedy decode.
- *Joint-loss / step-val / beam-3-BLEU* (seeds 1901–1908): corrects (iv) additionally—decoded-BLEU selection uses beam search (`eval_translation_beam_size=3`), matching the released config's evaluation metric. We include this variant because greedy-optimal checkpoints can differ from beam-3-optimal: the greedy "best" of the third protocol might be a checkpoint that beam-3 selection would not have chosen, so the fourth protocol removes this confound by selecting on the same beam-3 criterion the released model was trained under.

All four variants share the same architecture, data pipeline, manifests, and vocabularies (`config_sha256 0321104a...`); they differ only in loss composition, validation cadence, and selection decode. Table 5 reports the beam-3 dev BLEU-4 and PURE–REC gap for each. The corrected protocols confirm that the competence gap and the non-reproduction are not artefacts of the original-implementation differences: dev BLEU-4 remains 6–10.5 (vs. released 13.38) and PURE–REC gaps remain negative across all near-faithful and joint-loss runs.

3.5 Discovery timeline and pre-registration status

The +10.24 probe response was first observed on 2026-07-24 (canonical materialization; earlier pre-exclusion gap +11.01 on 2026-07-22). The six prospectively specified legacy seeds (101–606) were specified before their results were inspected; the eight extension seeds (707–1405) were added post-hoc. The Jaccard retriever, PURE protocol, and 641-query set were locked before reconstruction training began, after exploring the probes in Table 9. No external pre-registration exists; these six seeds constitute the only prospectively specified within-project follow-up evidence. Because the Jaccard pure-copy probe was selected as the maximal-gap probe post-hoc, its CI is descriptive, not prospectively confirmed. Full provenance in SI Sup. C.

Table 5: Training-protocol robustness. All variants share the released architecture and data; they differ only in loss composition, validation cadence, and selection decode. Each row reports only that row's runs (not aggregates across families). Dev BLEU-4 is the uniform beam-3 full-pool value; gap is the canonical PURE–REC sacreBLEU-4 difference on the 641-item test set. No variant meets the competence gate (dev BLEU-4 $\geq$ 12.38).

Protocol	Selection / validation	Dev BLEU-4	Gap range (neg. fraction)
Translation-only / epoch / NLL	NLL / epoch	6.5–9.4 (paper-derived)	$[-1.85, -0.80]$ (14/14 secondary)
+ validation-freq-misread	NLL / epoch	8.3–9.8	$[-1.70, -0.52]$ (4/4)
Translation-only / step-val / BLEu	greedy-BLEU / 14 steps	7.6–10.5	$[-1.37, -0.36]$ (10/10)
Joint-loss / step-val / greedy-BLEU	greedy-BLEU / 14 steps	6.3–10.2	$[-1.59, -0.48]$ (8/8)
Joint-loss / step-val / beam-3-BLEU	beam-3-BLEU / 14 steps	8.7–10.5	$[-1.38, -0.37]$ (8/8)
Released evaluator	—	13.38	+10.24

3.6 Metrics and uncertainty

The recorded test poses (REC) are decoded by each BT recognizer; we avoid the abbreviation "ground truth" because recorded poses are inputs to a learned measurement model, not a semantic gold standard. We report corpus BLEU-4 on the 0–100 sacreBLEU scale Post (2018) (sacrebleu 2.5.1 signature: `BLEU|nrefs:1|case:mixed|eff:no|tok:13a|smooth:exp|version:2.5.1`; max n-gram order 4; `effective_order=False`). BLEU-1, chrF, ROUGE-L, BERTScore Zhang et al. (2020), and WER (jiwer 3.1.0) provide cross-metric consistency checks (not independent semantic validation, as all share decoded outputs). BT likelihood provides a non-decoded control; pose-DTWp was withdrawn (see §4.1).

Uncertainty is quantified with donor-cluster bootstrap (primary; 10,000 resamples; clusters defined by donor identity, 565 unique donors across 641 queries, 61 donors reused with maximum reuse 6, Gini 0.11; SI Sup. G) and item-level bootstrap (SI Sup. G). A pairwise difference is called detectable when its interval excludes zero. For seed dispersion we report *descriptive* Student-t intervals over the 14 paper-derived seeds (six primary + eight extension) and over the 10 step-corrected seeds; these intervals assume same-implementation seed outcomes are approximately IID/normal, an assumption that leave-one-out sensitivity (SI Sup. B) checks for robustness but cannot validate. They should be read as "given a future random seed under this identical implementation" dispersion summaries, not as calibrated population-level prediction intervals or independent replication statistics; raw seed values are listed in Table 11 and SI Tables S6–S7. The six primary seeds give a descriptive interval $[-2.48, -0.28]$; all 14 leave-one-seed-out folds keep the interval below zero (SI Sup. B). The headline gap is a **post-selected descriptive estimate** (§4.1); all other contrasts are Holm-corrected within a defined family or explicitly labelled descriptive (SI Sup. C).

3.7 Terminology: run, checkpoint, unique binary, family, evaluator

Throughout, *run* denotes a single training execution; *checkpoint* denotes the selected model weights from that run; *unique weight binary* counts distinct SHA-256 hashes of checkpoint files (the cf_202 and ls_202 runs produced byte-identical weights from the same wandb run); *family* denotes a group of runs sharing a training protocol; and *evaluator* denotes a checkpoint used to decode fixed outputs. The 14 paper-derived reconstruction runs use NLL selection with epoch-level validation; the 4 validation-freq-misread runs use NLL selection with epoch-level validation (the validation cadence was initially misread); the 10 step-corrected runs use decoded-BLEU selection on step-level validation (every 14 optimiser steps, matching the released `validation_freq=14` semantics); the joint-loss variants additionally add CTC recognition loss. All share the documented architecture and hyperparameters (§3).

Custom-terminology registry.

Table 6 maps each author-defined term to its estimand, dataset, and evidence grade. Evidence grade: D = descriptive (no formal error control); all custom quantities in this paper are descriptive.

Table 6: Custom-terminology registry. Each term maps to its estimand, the dataset on which it is computed, and its evidence grade (all descriptive).

Term	Estimand	Dataset	Grade
Training-pool read-out	In-sample autoregressive free-decode BLEU/EM	Full 7,060-item training set	D
PURE–REC gap	Corpus sacreBLEU-4 difference (PURE − REC)	641-item PHX-public test	D (post-selection)
Operational tolerance gate	One-sided: dev BLEU-4 $\geq$ 12.38 *and* dev WER $\leq$ 86.37	515-item dev set	D

Claim hierarchy.

We classify every claim in this paper into four evidence tiers:

- *Descriptive (D)*: observed pattern with no formal error control. Most findings in this audit are descriptive.
- *Sensitivity (S)*: the result is qualified by an explicit post-hoc condition (e.g., single-annotator, matched subset, post-selection).
- *Prospectively specified within-project follow-up (C)*: pre-specified non-reproduction direction. Only the six legacy seeds (101–606) qualify; these seeds share a single implementation and data pipeline, so this is seed-level variation within one implementation, not six independent replications.
- *Not identified (NI)*: the claim cannot be made from available data. The causal checkpoint-identity effect is NI because competence is confounded.

Table 7: Claim hierarchy. Each key claim mapped to its evidence tier and supporting evidence.

Claim	Tier	Evidence
Our disclosed reconstruction did not attain released evaluator competence	D	28 recipe-conditioned runs, all dev BLEU < 10.5; not a public-artifact-insufficiency claim
Probe response does not reproduce across weaker checkpoints	D	61/61 non-degenerate gaps negative
Probe response reflects a causal checkpoint-identity effect	NI	Competence confound: 6.5–10.5 vs 13.38
Headline gap +10.24 is post-selection	S	Max-gap probe among 8 explored; CI descriptive
Reference replacement attenuates the gap (90.5%)	S	Single-annotator, 461-item, 4 confounded changes
Origin effect persists under matched pools (+9.56)	S	148 items, SMD<0.1; association not causal
Probe is adversarial lexical-coupling stress test	D	Source = reference in PHX-public; retriever uses source Jaccard

Experimental stratification.

We classify every checkpoint family into three evidential tiers (a fourth, holdout controls, was planned but the corresponding runs were not recoverable from disk after a host compromise and are omitted):

- *Near-faithful (corrected-recipe)*: 10 step-corrected + 4 validation-freq-misread = 14 runs. The 10 step-corrected runs match the released validation schedule exactly (step-level validation every 14 optimiser steps, decoded-BLEU selection); the 4 validation-freq-misread runs initially read `validation_freq=14` as 14 epochs before the step-corrected correction. These are the closest recipe matches and the primary recipe-faithfulness evidence for the reconstruction-failure claim.
- *Secondary (legacy-implementation replication)*: 14 paper-derived seeds (6 prospectively specified seeds 101–606 + 8 extension seeds 707–1405; legacy script, 300 epochs, NLL selection). The 6 prospectively specified seeds are the only *prospectively specified* within-project follow-up evidence (pre-specified non-reproduction direction); the family shares a known deviation from the released protocol (legacy script, NLL selection) and serves as supplementary stability evidence.
- *Diagnostic (post-hoc sensitivity)*: 8 joint-loss greedy + 8 joint-loss beam-3 + 2 confirmation + 1 rescue + 15 distillation + 4 ladder + 1 long-schedule + 1 joint-loss supplementary = 40 runs. These probe mechanism, protocol robustness, and competence extrapolation.

The reconstruction-failure claim rests on the 14 near-faithful runs (10 step-corrected, 4 validation-freq-misread; descriptive step-corrected 10-seed gap range $[-1.37, -0.36]$, all 10 negative) together with all 14 legacy seeds; the *prospectively specified* non-reproduction direction comes only from the 6 prospectively specified legacy seeds (101–606, prediction interval $[-2.48, -0.28]$, all six leave-one-seed-out folds below zero). Diagnostic runs serve as triangulating context and must not be cumulated with prospectively specified within-project evidence to inflate the appearance of independent replication. Table 8 gives the full accounting, generated from a single machine-readable registry (`results/accounting_table.json`).

4 Results

4.1 Adversarial probe response and conditioning controls

Under the released SLRTP2025 evaluator, source-conditioned whole-donor retrieval scored 23.02 sacreBLEU (0–100 scale), compared with 12.78 for the recorded test poses—a gap of +10.24 BLEU points (95% CI $[+8.87, +11.64]$; 10,000 donor-cluster resamples; item-level bootstrap gives the nearly identical $[8.88, 11.62]$). This is the expected output of a deliberately adversarial lexical-coupling stress test (§3.2): the retriever selects donors by surface vocabulary overlap with the source, which is identical to the scoring reference, so the gap measures the evaluator's susceptibility to this specific shortcut rather than general retrieval quality or signing adequacy. A deployment-realistic noisy-gloss retrieval pipeline scores only -0.13 relative to REC under the same evaluator (§4.5), confirming that the gap is specific to the constructed adversarial probe.

Controls localized the observation to source-conditioned donor retrieval (SI Table S22): pure donor copy (23.02) exceeded PT-composed (18.86), oracle-gloss (11.4–11.8, a leakage diagnostic), PT baseline (1.59), and 20-seed random-donor baselines (0.90/0.77). Source-text Jaccard was the strongest retriever (vs. LaBSE 19.2, multi-SBERT 16.5). Non-decoded controls do not reproduce the gap: teacher-forced NLL gives REC 3.060 vs PURE 3.479. We do *not* report an item-level pose-DTWp here: a well-defined pose-distance protocol (alignment, per-frame normalization, missing-frame handling, direction convention) is beyond the scope of this audit, and we withdraw the earlier "near-zero" remark that lacked a computed backing; the only pose-level metric we report is the leaderboard DTW-MJE (SI Table S25). Complementary text metrics (BLEU-1, chrF, ROUGE-L, WER, BERTScore) agree on direction (SI Table S22); decoding-parameter sweeps (beam 1/3/5, length penalty, max length) leave the gap at +9.9 to +10.8 (SI Sup. F).

Probe multiplicity and post-selection status of the headline gap.

Before freezing the headline probe, we explored four retrieval strategies (source-text Jaccard, LaBSE, multi-SBERT, oracle gloss) crossed with three construction variants (pure copy, PT-composed, oracle-gloss-composed) on the released evaluator alone. Of the 12 retrieval–construction cells, 6 were explored, 3 are not applicable (oracle-gloss-composed is definitionally tied to the oracle-gloss retriever, so it cannot be applied to Jaccard/LaBSE/multi-SBERT without changing the retriever), and 3 were not explored (LaBSE×PT-composed and multi-SBERT×PT-composed

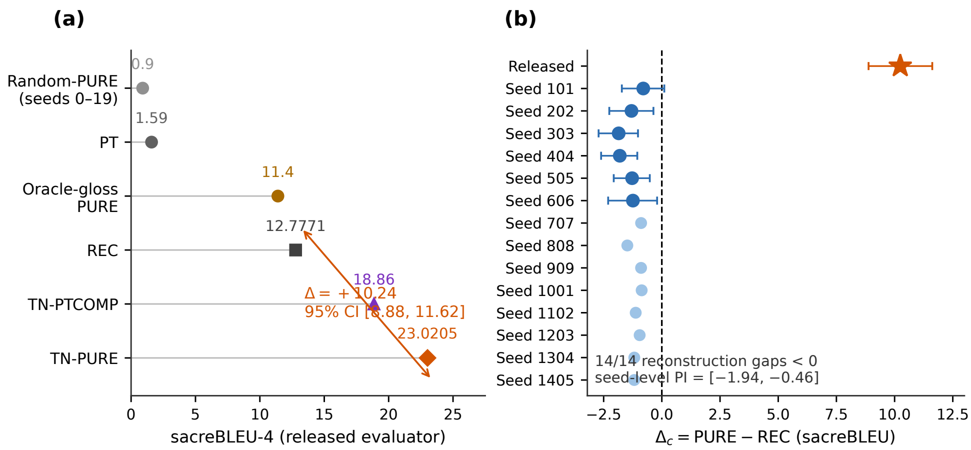

Fig. 1: Headline reversal under the released evaluator and its absence across recipe reconstructions. (a) System scores under the released evaluator (sacreBLEU-4, canonical 641-query protocol): TN-PURE 23.02 vs. REC 12.78 ($\Delta = +10.24$, post-selected descriptive 95% CI $[8.87, 11.64]$), with TN-PTCOMP 18.86, PT 1.59, oracle-gloss PURE 11.4, and random-donor PURE 0.90 (seeds 0–19, ± 1 SD). (b) Per-checkpoint PURE$-$REC gaps: released evaluator (+10.24) vs. fourteen *paper-derived* (legacy) recipe reconstructions (seeds 101–1405), all 14 negative (donor-cluster bootstrap 95% CIs; descriptive seed dispersion range $[-1.85, -0.80]$, all 14 values below zero). The 10 step-corrected and 16 joint-loss families are shown in Fig. 2; none of them produces a positive gap either (see Table 8). All reconstructions are less competent than the released evaluator; the figure demonstrates non-replication within the constructible family, not an isolated causal checkpoint effect.

were deemed low priority after Jaccard pure-copy dominated; oracle-gloss×oracle-gloss-composed is subsumed by oracle-gloss×PT-composed). We additionally explored two random-donor baselines that lie outside the 4×3 matrix (Table 9 and `results/probe_multiplicity.json`). The headline Jaccard pure-copy probe was selected as the *maximal-gap* cell among the explored cells; its 95% CI $[+8.87, +11.64]$ is therefore **post-selection descriptive**. We do not apply a multiplicity correction: the family of explored probes was defined after observing the gaps, so no correction would make the result confirmatory. The six prospectively specified legacy seeds (§3.5) were pre-specified and provide the only prospectively specified evidence (for the non-reproduction direction, not for the gap magnitude).

The probe reversal motivates the prior question: can the public recipe reproduce the released evaluator's competence at all?

4.2 RQ1: Failed public-artifact reconstruction—what the released evaluator has that reconstructions lack

The non-reproduction in RQ2 raises the prior question of RQ1: can the public recipe reproduce the released evaluator's competence? We approach it through three descriptive comparisons (Fig. 2b,c; Table 10; full version in SI Sup. D), reporting them as descriptive triangulation, not three independent tests.

Readout* with *generalization: the distinctive combination.

The released evaluator decodes its full 7,060-item training pool at free-decode BLEU 78.8 with 70.7% exact-match while retaining dev/test performance of 13.38/12.78. The constructible family stays far below: reconstructions span train BLEU 7.2–9.3 at dev BLEU 6.5–9.4, and distillation students 6.6–8.9 at dev 7.0–9.0. Across 31 checkpoints with uniform full-pool readouts (30 constructible plus released), training-pool readout tracks decoded competence (Spearman ρ=0.916, n=31; ρ=0.907 with the released evaluator excluded, n=30, confirming the correlation is not driven by the released extreme point) and is uncorrelated with the overfit indicator (ρ=0.040) or the PURE–REC gap (ρ=−0.082; SI Sup. D; full robustness breakdown in `results/robustness_diagnostics.json`). The released evaluator's readout-with-generalization combination is not explained by any within-family overfit–readout relationship.

Leakage sanity checks for the train-pool readout.

The 78.8/70.7% readout is not a trivial implementation artifact: five controls (SI Sup. M) rule out four shortcut classes — teacher forcing, output–reference misalignment (label permutation collapses BLEU from 78.6 to ∼1.0), pose-ignorant ID/cache lookup (input-side pose permutation collapses BLEU to <1; a same-signer held-out control reaches only 12.19 BLEU / 3.2% EM), and narrow-text memorization. The checks do not identify the training mechanism but are consistent with pose-conditioned training-pool exposure.

Distillation and competence extrapolation.

Fifteen distillation students ($\alpha \in \{0, 0.25, 0.5, 0.75, 1.0\}$, three seeds each) all have negative gaps and do not transfer the readout pattern (SI Table S26). Competence extrapolation (OLS/GP), per-item gap decomposition, and Gaussian-weight-noise perturbation are reported in SI Sup. K and F; neither resolves the identification problem inside the unobserved $(10.5, 13.38)$ interval. The reversal is locally robust near the released checkpoint (Gaussian noise $\sigma \leq 3\times10^{-3}$ preserves it within ±0.75; 9 released-weight fine-tunes all pass the competence gate), but these probes test the released checkpoint's vicinity, not an independent construction.

Synthesis.

Across the three descriptive comparisons, the released evaluator is the only checkpoint combining high training-pool readout (78.8 BLEU, 70.7% EM) with retained in-domain performance (13.38/12.78). No reconstruction or distilled student reproduces either the +10.24 gap or this combination.

4.3 RQ2: Multi-checkpoint reproduction of the stress-test ordering

TN-PURE-v1 yields an original-evaluator gap of +10.24 BLEU points (95% CI $[+8.87, +11.64]$) and TN-PTCOMP-v1 +6.09 $[+4.76, +7.41]$; across all fourteen reconstructions both gaps are negative (PURE range $[-1.85, -0.80]$; seed-level 95% prediction interval $[-1.94, -0.46]$, mean -1.20, SD 0.33, 0/14 positive; the six prospectively specified legacy seeds alone give $[-2.48, -0.28]$ and the eight extension seeds $[-1.61, -0.54]$, with the interval below zero in all fourteen leave-one-seed-out folds), and for the six prospectively specified legacy seeds, original-minus-run difference-in-differences estimates range from +10.89 to +11.99, every interval excluding zero.

4.3.1 Recipe-conditioned evaluator reconstructions

All fourteen reconstructions share frozen manifests, architecture, hyperparameters, and selection rule; seeds 101–1405 vary initialization and data order (Table 11), and all evaluators are decoded with the canonical §3.2 donor registry (exact-normalized-text exclusion applied).

Competence is assessed on the released 515-sequence development subset, never on PHX-public. The released evaluator obtains dev BLEU-4 13.38; under the uniform full-pool beam-3 free-decode protocol the 14 paper-derived reconstructions obtain dev BLEU-4 6.5–9.4, and the 14 near-faithful runs obtain 7.6–10.5 (with joint-loss beam-3 contributing the upper end). Their selected validation NLL (2.641–2.712) is lower than the released evaluator's teacher-forced dev NLL (3.235), so likelihood and decoded recognition competence are not interchangeable. Consequently, competence and unavailable original-training details remain confounded with evaluator identity; this is a descriptive observation, not a causal checkpoint-identification result.

Competence gate and trajectory analysis.

A checkpoint is competence-matched only if both dev BLEU-4 ≥ 12.38 and dev WER ≤ 86.37 (one-sided tolerance around released 13.38 / 83.37). None of the 52 gate-eligible non-distillation training runs passes the gate. The 9 released-weight fine-tunes (SI Sup. F) all pass (dev BLEU-4 12.92–13.18), but they perturb the released weights rather than constructing from the recipe. The rescue search's best variant (weight-decay 0, seed 202) yields gap -1.59. The gate conclusion is threshold-insensitive across a $\pm[0.5, 2.0]$ BLEU $\times$ $\pm[1.5, 4.5]$ WER grid. Full details in SI Sup. D–E.

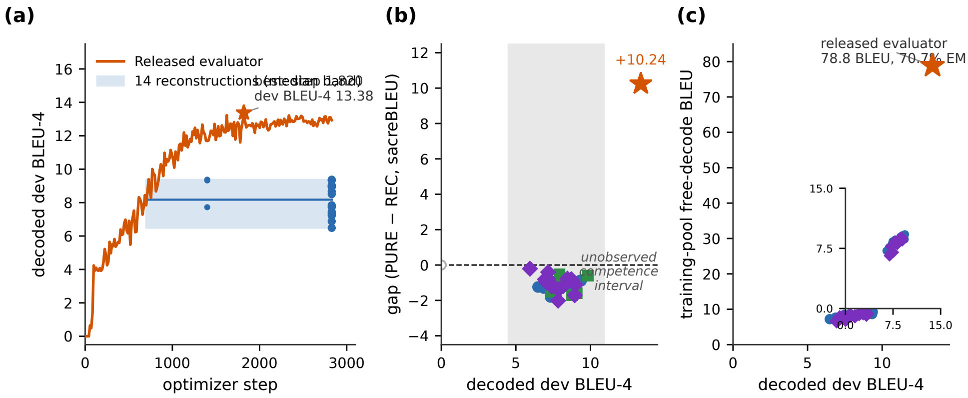

Fig. 2: Decoded competence and training-pool readout separate the released evaluator from every constructible checkpoint. (a) Dev-BLEU-4 training trajectories on a common optimizer-step axis: the released evaluator (orange) reaches 13.38 at step 1,820 of 2,828 (archived validation log, 202 logged points); fourteen reconstructions form a median band (blue), plus uniformly decoded epoch-25/50 points. (b) Canonical PURE−REC gap vs. decoded dev BLEU-4 by family: reconstructions (circles), validation-freq-misread/step-corrected/confirmation/long-schedule (squares), joint-loss greedy/beam-3 (triangles-up/triangles-down), rescue (triangle-right), distillation students (diamonds), released evaluator (star); the gray band marks the observed 6.5–10.5 range (uniform protocol) and the (10.5, 13.38) interval is unobserved by recipe-constructed checkpoints (the nine released-weight fine-tunes occupy 12.92–13.18 but are weight perturbations); degenerate checkpoints are hollow gray points. (c) Full-training-pool free-decode readout (beam-3, full 7,060 poses) vs. decoded dev BLEU-4: every constructible checkpoint reads out 6.6–9.3 BLEU, while the released evaluator reads out 78.8 BLEU (70.7% exact match) at dev 13.38. Families are descriptive; checkpoints are not IID replicates.

What the released training log establishes.

The bundle's `validations.txt` logs dev BLEU-4 reaching 13.38 at step 1,820 (Fig. 2a), while all constructible reconstructions plateau at 8–10 by epoch 50–100; the training dynamic behind this slope is not explained by public artifacts. Input degradation (temporal downsampling) compresses both scores but does not eliminate the reversal (at k=8: REC 8.75, PURE 11.69, gap +2.94; SI Sup. F).

Cross-metric consistency (SI Table S22).

Under the original evaluator, the PURE–REC gap is positive and large for all decoded-text metrics (BLEU-1 +20.94, chrF +12.93, ROUGE-L +17.87, WER +13.90 sign-flipped so positive means PURE has lower error than REC; sign conventions in SI Table S22 footnote), while teacher-forced NLL is more negative for PURE (−0.419, also sign-flipped), consistent with the likelihood-vs-decoded-metric divergence Jiang et al. (2025). Under all six prospectively specified legacy reconstructions the gap is negative across every metric. The probe response is therefore a property of decoded-text metrics under the original evaluator, not specific to sacreBLEU-4. Full table in SI Table S22.

Rank stability (SI Sup. H).

On a fixed nine-system stress-test set, the released evaluator ranks TN-PURE first; five of six reconstructions rank REC first, with moderate cross-checkpoint agreement (mean τ_b=0.44, ρ=0.57).

Reference-frame sensitivity: matched-subset analysis.

Two 2026 corpus audits bear directly on this reversal. Czehmann et al. Czehmann et al. (2026) released manual sign-to-text back-translations of the test set (one deaf fluent L2 DGS signer; per-item confidence flags), documenting information loss, lexical errors, and translationese in the original speech-transcribed references, and showed that reference replacement already changes BLEU scores and rankings on PHOENIX-2014T. Alkain et al. Alkain et al. (2026) quantified train–test text overlap across sign-language translation corpora, showing that a small fraction of highly training-like test items disproportionately inflates corpus BLEU. We reproduce the overlap pattern locally: the 32 test items whose reference exactly matches a training text score REC 40.26 / PURE 78.73 (gap +38.5), while the remaining 609 items score REC 11.93 / PURE 22.20 (gap +10.3).

Czehmann et al. compared original- and human-reference conditions on the same confidence=1 subset (462 of 642 items); we follow the same protocol. On the matched 461-item subset (after join with our 641 released test IDs), the original-reference gap is +10.03 BLEU points (REC 14.94 / PURE 24.98) and the human-reference gap is +0.95 (REC 7.35 / PURE 8.30): a paired attenuation of 9.08 BLEU points (90.5%, bootstrap CI $[7.44, 10.86]$ on the absolute attenuation and $[0.81, 1.01]$ on the attenuation ratio; SI Table S31; SI Fig. S5; per-item paired view in SI Fig. S2). The interval for the residual gap includes zero (paired item CI $[-0.13, 1.99]$). This 90.5% is a *single-annotator, 461-item, post-hoc point estimate*; it cannot isolate any single mechanism because reference replacement simultaneously changes (i) reference content, (ii) expression style/translationese, (iii) BLEU floor level, and (iv) the lexical coupling between retrieval selection and scoring reference. Teacher-forced BT likelihood does not reproduce the probe response under either reference (REC NLL 3.060 vs PURE 3.479 on original refs; 3.41 vs 3.71 on human refs) Jiang et al. (2025).

Scoring against both references jointly leaves the gap at +11.50, *larger* than original: adding the original reference injects a second lexically coupled target. The probe response is thus specific to reference frames containing the original speech-transcribed register. This is a descriptive contrast, not a causal attribution. Under all six reconstruction checkpoints the human-reference gap is negative (-0.15 to -0.65), and the asymmetry is consistent across metrics (chrF 82.2%, BLEU-1 84.0%; SI Table S23). The asymmetric PURE-vs-REC drop (under reference replacement REC drops 50.8%, PURE drops 66.8%, a 1.31× relative-drop ratio; the absolute-drop ratio is 16.68/7.59 = 2.20) is numerically inconsistent with simple symmetric floor compression (SI Table S23), though BLEU's non-linearity precludes a definitive test.

The human back-translations are sign-conditioned paraphrases of the recorded test poses, not judgments of the replay donors, so semantic adequacy of TN-PURE-v1 is left unassessed.

4.4 Donor-origin contrast

If the released-evaluator probe response requires exposure to training-pool donors, retrieval restricted to unseen poses should remove or invert it: under the original evaluator, UNSEEN-PURE-v1 (retrieving from the 640 other test poses) scores 8.51 sacreBLEU vs. REC 12.78 (gap −4.27), a donor-pool substitution estimand of +14.51 BLEU points (SI Table S30). This is a *pool-substitution difference*, not a pure exposure contrast: SEEN and UNSEEN differ simultaneously in pool size (7,060 vs 640), median donor–reference Jaccard (0.50 vs 0.37), donor reuse (Gini 0.11 vs 0.29), and training-pool exposure.

Matched donor-pool analysis (SMD<0.1 control).

To separate the donor-similarity confound from the pool-origin contrast, we computed a per-item matched subset restricting to items where the seen-donor and unseen-donor Jaccard similarities differ by at most 0.05. This yields 148 items with SMD=0.065 on Jaccard (vs. +0.554 on the full 641). On this matched subset, the pool-origin contrast is +9.56 BLEU [+7.47, +11.76]—still large with CI excluding zero. We do *not* interpret the change from the unmatched +14.51 as a percentage attribution: the two estimates come from different target populations (n=641 vs n=148), and the caliper selection shifts sentence-length, template-density, and difficulty distributions, so corpus-BLEU differences between them are not additive. This is a Jaccard-balanced selected-subset association, not an identified training-exposure effect: the matched analysis controls for donor-reference Jaccard but not for pool size (7,060 vs 640) or the diversity of available donors. Full results in `results/matched_donor_pool.json`; additional balance checks in SI Sup. E.

Pool-size-controlled resampling.

A stronger design equalizes pool size directly. We repeatedly subsample 640-item donor pools from the 7,060-item training pool (matching the 640-item test pool), apply the same source-Jaccard retrieval rule, and score using the released evaluator's cached training-pool decoded hypotheses (no re-decoding). Over 1,000 resamples, the pool-origin contrast is +8.38 BLEU (median +8.39, 95% conditional sampling interval [+7.44, +9.34], fraction positive 1.00), reduced from the full-pool +14.51 but still large with the interval excluding zero. Pool size and the diversity of the available donors thus account for part of the unmatched difference, but a substantial pool-origin contrast persists even at equal pool size. This is still an association—training exposure, donor reuse, and donor diversity remain confounded with pool origin—but it removes the pool-size confound that neither the unmatched nor the matched-subset estimate could. Full distribution in `results/donor_pool_resampling.json`.

Balanced factorial and path decomposition.

The 605-query factorial (matching seen/unseen donors at high/low LaBSE similarity; SMD +0.4 within strata) estimates the pool-origin main effect at +2.08 $[+1.56, +2.57]$, similarity at +6.96 $[+5.93, +8.08]$, and interaction at +3.70 $[2.63, +4.79]$ (Holm p=0.0003); across reconstructions the seen main effect shrinks to +0.51. Because seen donors remain more similar than unseen within strata, this is a provenance contrast, not an identified training-exposure effect. The matched-subset analysis above provides a *differently conditioned association* (caliper-restricted n=148 vs. factorial n=605 vs. full n=641, three different target populations); the factorial is retained for its 2×2 decomposition.

4.5 Competitive-system context

Because the audit's adversarial probe is constructed rather than competitive, the public SLRTP2025 leaderboard offers external context: the "Ground Truth" row reproduces our canonical REC numbers exactly (BLEU-4 12.78, Walsh et al. (2025); SI Table S25). The retrieval-based winner USTC-MoE sits roughly even with REC on lexical BT metrics (BLEU-4 −0.72; SI Table S25). We re-decoded two realistic retrieval outputs under all available evaluators (SI Sup. P): the deployment-realistic noisy-gloss retrieval pipeline decodes at 12.65 under the released evaluator (gap −0.13 vs. REC), and our rebuild of the winner's fixed outputs from its released intermediates decodes at 2.42 (gap −9.94; a failed reproduction, not representative of the original entry). Under all 62 non-degenerate reconstructed evaluators, both systems give negative gaps (noisy-gloss range $[-2.17, -0.07]$; USTC-MoE range $[-9.52, -4.28]$). Neither realistic system recovers a retrieval-favoring ordering under any evaluator. The +10.24 gap is specific to the constructed adversarial probe, and we do not claim that real competition rankings would change across checkpoints. Archiving fixed per-team outputs alongside public scores is the most direct suggestion we can offer for enabling rank-stability audits.

Per-item gap decomposition (experiment C, SI Sup. K).

An OLS regression ($R^2 = 0.33$, $n = 641$) attributes the largest share of per-item gap variance to high-overlap-train count, template density, and donor–reference Levenshtein distance; removing the signer block changes R^2 by $\leq 1\%$ (ΔR^2). The response variable is the per-item PURE-minus-REC BLEU difference computed by aggregating the per-item n-gram sufficient statistics that sacrebleu maintains internally (this gives an exact per-item BLEU-4 contribution under the corpus-level exp-smoothing); SEs are heteroskedasticity-robust (HC3) and also reported with two-way clustering on signer and broadcast show as a sensitivity check. The model reported is the AIC-minimal specification across a small grid of additive and pairwise-interaction candidate regressors selected before fitting; we do not claim it is the unique correct model. Full coefficient table, alternative-model sensitivity, and figure in SI Sup. K.

5 Discussion

The audit yields three findings organized around the question of whether, using our disclosed reconstruction implementation and the publicly documented recipe, we can attain the released SLRTP2025 BT evaluator's competence and reproduce its behavior on a constructed replay probe. We emphasise that this is a stricter question than "are the public materials themselves sufficient?"—the latter would require an independent reference implementation whose correctness is independently validated, which we could not obtain (the released repository is eval-only; SI Sup. R).

RQ1—Failed reconstruction by our disclosed implementation. Using our disclosed reconstruction implementation and the publicly released artifacts, none of the 14 near-faithful runs (4 validation-freq-misread, 10 step-corrected) or 14 legacy-implementation seeds reached the released evaluator's competence on the *primary continuous metrics*: dev BLEU-4 6.5–10.5 vs. 13.38, training-pool readout 7.2–9.3 vs. 78.8, and teacher-forced token accuracy 55–60% vs. 96.5% (Table 4). Three descriptive comparisons (Table 10) all place the released evaluator far outside any constructible range on these continuous metrics; the binary competence gate (dev BLEU-4 $\geq$ 12.38, dev WER $\leq$ 86.37; §3.1) is a sensitivity summary of these gaps, not a public-sufficiency judgment. The source of the gap remains unidentified; it may reflect information incompleteness in the public artifacts, stochastic underspecification, or a residual defect in our disclosed implementation (the four protocol mismatches documented in §3.4 illustrate how the latter category was identified and partially corrected across rounds). A protocol-robustness check confirms that correcting the four original-implementation differences (joint CTC+translation loss, step-level validation, decoded-BLEU selection) does not close the competence gap or reproduce the probe: the corrected joint-loss protocols also plateau at dev BLEU-4 6–10.5 with strictly negative PURE–REC gaps.

RQ2—Descriptive non-reproduction under competence-unmatched reconstructions. Because every constructible checkpoint is less competent than the released evaluator (dev BLEU-4 6.5–10.5 vs. 13.38; decoded family max 10.5 from joint-loss beam-3, near-faithful max 10.5 from step-corrected), the non-reproduction of the adversarial probe response (all 61 of 61 non-degenerate gaps negative, range $[-2.01, -0.21]$) *cannot distinguish* whether it reflects the competence gap, an unrecovered training mechanism, or their interaction. This is a descriptive observation about our public-artifact reconstruction attempt, not a causal checkpoint-identity claim. Without a competence-matched independent implementation—which we could not obtain (the released repo is eval-only; our trained checkpoints were lost in a host compromise)—the identification problem is unresolved. Two local-robustness checks on the released checkpoint (Gaussian weight noise and fine-tuning from released weights, SI Sup. F) confirm the reversal is locally robust but probe only the vicinity of the released weights; competence-extrapolation and per-item decomposition analyses (SI Sup. K) are exploratory and do not resolve the identification problem. The descriptive step-corrected 10-seed gap range $[-1.37, -0.36]$ is entirely below zero, and the prospectively specified 6-seed legacy interval $[-2.48, -0.28]$ does not include zero; the prospectively specified non-reproduction direction rests on the latter. Cross-evaluator re-decoding of a realistic noisy-gloss retrieval output yields a released-evaluator gap of -0.13 and 62/62 negative gaps under non-degenerate reconstructed evaluators (range $[-2.17, -0.07]$), confirming the probe response is specific to the constructed adversarial design; our rebuilt USTC-MoE output (SI Table S33) is pipeline-faithful but not score-faithful (2.42 vs. official 12.06) and should not be taken as representative of the original competition entry.

RQ3—Reference-frame sensitivity and donor-origin coupling. On the matched 461-item confidence=1 subset of Czehmann et al.'s Czehmann et al. (2026) human back-translations, reference replacement is associated with a gap reduction to +0.95 from +10.03—a single-annotator, 461-item, post-hoc point estimate of 90.5% attenuation. This estimate cannot isolate any single mechanism because reference replacement simultaneously changes (i) reference content (speech transcription vs. sign-conditioned back-translation), (ii) expression style and translationese, (iii) BLEU floor level (REC drops 7.59, PURE drops 16.68), and (iv) the lexical coupling between retrieval selection and the scoring reference. The asymmetric PURE-vs-REC drop (relative-drop ratio 1.31×: REC drops 50.8%, PURE drops 66.8%; absolute-drop ratio 2.20×) is numerically inconsistent with simple symmetric floor compression, but BLEU's non-linearity and the coupling structure preclude a definitive floor-effect test. The matched donor-pool analysis (§4.4) shows the pool-origin contrast persists at +9.56 under SMD<0.1 Jaccard matching (148 items); the pool-size-controlled resampling (640-vs-640, §4.4) gives +8.38 $[+7.44, +9.34]$. We report these as Jaccard-balanced and pool-size-controlled associations, not percentage attributions. The pool-substitution reverses the gap sign (UNSEEN-PURE-v1 scores 8.51 vs. REC 12.78, gap −4.27); all three pool-origin-contrast estimates remain associations, not identified training-exposure effects, because donor diversity, reuse, and training exposure are not independently manipulable.

Relation to prior work.

These findings extend prior critiques of BT validity from output faithfulness to public-recipe reproducibility: improved BT scores need not imply target conditioning Hong and Košecká, Jana (2026), low recorded-pose BT baselines are documented Walsh et al. (2024), and BT likelihood is a more stable correlate of human judgment than decoded BLEU Jiang et al. (2025). Our reversal is a *decoded-sacreBLEU* reversal — teacher-forced BT likelihood does not reproduce it (REC NLL 3.060 vs PURE 3.479) — and whole-donor replay is a constructed probe, whereas SOKE-style word-level retrieval can improve pose quality under a different representation Zuo et al. (2025).

Scope limitation: our reconstruction vs. public-artifact sufficiency.

Our conclusion is strictly about our disclosed reconstruction implementation: using the public artifacts as we interpreted them, no construction we tested attained the released evaluator's competence. The stronger claim—"the public materials themselves are insufficient"—would require an independent reference implementation whose correctness is independently validated (e.g., by reproducing some other known-good checkpoint from the same recipe family, or by the original authors releasing the training command). We do not have such validation: the released SLRTP2025 repository is eval-only (no `train.py`), our request for the original training command went unanswered, and our four protocol mismatches (§3.4) illustrate how easily an inference can be wrong. We therefore report the gap as descriptive evidence about our reconstruction, and treat the public-sufficiency question as unresolved pending either an authoritative training command or an independent reconstruction that does attain the released competence. Under that scope, the absence of reproduction across 11 protocol variants (translation-only, step-corrected, joint-loss greedy, joint-loss beam-3) and 68 from-scratch runs is strong evidence about the difficulty of reconstructing this evaluator from public artifacts, but stops short of proving impossibility.

Value, finding, and conclusion reproducibility.

Following the tripartite framework of Cohen et al. (2018), the audit supports a partial and asymmetric reproducibility claim. *Value* (recomputing the headline numbers from the released artifacts) is reproduced: the REC BLEU-4 of 12.78, PURE of 23.02, and gap of $+10.24$ are regenerable from the canonical donor registry and are stable across pre-exclusion materializations (§3). *Findings* (the probe ordering and training-pool readout pattern) do *not* reproduce under any of the 61 constructible non-degenerate checkpoints, all of which are less competent than the released evaluator. *Conclusions* (the recipe-sufficiency claim that, under our disclosed reconstruction, the public artifacts are adequate to recover the evaluator's behavior) are not supported by our data: the unobserved competence interval $(10.5, 13.38)$ lies above every constructible checkpoint ceiling, and the step-corrected gap range $[-1.37, -0.36]$ is entirely below zero (i.e., the recipe-constructed non-reproduction direction is unambiguous within its family). This separation is important because it localises the reproducibility failure observed in our reconstruction to the training-recipe interpretation rather than to a computational or scoring error, while making explicit that an unidentified training mechanism, a competence-gated effect, or a defect in our disclosed implementation remains possible in the unobserved interval.

Value of the audit infrastructure despite the competence confound.

The competence-matched condition is unmet, so RQ2's non-reproduction cannot be attributed to a causal checkpoint effect. This constrains what the audit *infers* but not what it *provides*: (i) the provenance table (Table 2) and three-way data reconciliation establish the released materialization differs from the official split by 41 IDs; (ii) the checkpoint registry with SHA-256 deduplication (SI Sup. D) prevents double-counting; (iii) the unified-inference regression test (§3.4) rules out several basic plumbing failures (data loading, masking, decoding alignment, vocabulary mismatch) but does not rule out subtler training-objective or scheduler-timing defects (see scope limitation, §5); (iv) the leakage controls (SI Sup. M) are extensible to any learned evaluator; (v) the single canonical materialization with four-layer reproduction (§3) is a concrete model for reproducible benchmarking infrastructure. These contributions are transferable to future learned-evaluator audits.

Actionable recommendations for benchmark organizers.

The audit yields three transferable lessons, scoped to the constructed replay-probe stress case on PHX-public rather than to general SLP leaderboard stability.

Lesson 1: a released evaluator is a research artifact, not an infrastructure component. The released `config.yaml` alone was insufficient for 68 reconstruction runs across 11 families to reach competence parity, and the readout pattern could not be localized to a training stage because intermediate checkpoints are unpublished. Benchmarks should release (i) the exact training command and seeds, (ii) data pipeline scripts beyond the manifest, (iii) a SHA-256 manifest of training tensors, (iv) intermediate checkpoints at least every 100 epochs, and (v) the fixed per-team pose outputs alongside the public scores; the current release provides only the Progressive Transformer baseline output, so rank-stability audits across independently trained evaluators cannot be reproduced by third parties on real competition systems.

Lesson 2: on template-dense benchmarks, retrieval entries may benefit from donor-exclusion disclosure. A donor-exclusion sweep is cheap and exposes the reference-coupling gap on PHX-public's template-dense register; retrieval-based leaderboard entries *may* voluntarily attach $\tau \in \{0.3, 0.5\}$ donor-exclusion results as a transparency practice on template-dense benchmarks. The training-pool free-decode BLEU ratio (released: $78.8 / 13.38 \approx 5.9$; reconstructions: 0.93–1.12) and input-side pose permutation with output-equivariance verification (SI Sup. M) are two descriptive probes we found informative in this case study; whether they generalize to other benchmarks is untested.

Lesson 3: reference-coupling artifacts warrant a supplementary reference check. On PHX-public, the 90.5% attenuation under Czehmann et al. Czehmann et al. (2026) back-translations is a single-annotator reference-frame case study showing that the gap is sensitive to the reference frame; it should not be generalized beyond this signer's interpretation. Benchmarks on similarly template-dense corpora could adopt human back-translations as a supplementary reference set, follow the matched-subset protocol, and flag systems whose gap shifts between the two; the cost is one reference file and a paired bootstrap. Whether this generalizes beyond PHX-public's lexical-coupling structure is untested.

Limitations.

The audit is confined to PHX-public, a template-dense weather domain with documented reference-validity and train–test-overlap issues Czehmann et al. (2026); Alkain et al. (2026). Per-team pose outputs are not archived; our cross-evaluator studies decode at or below REC under all 61 non-degenerate evaluators. All constructible checkpoints decoded on PHX-public (61 non-degenerate; Table 8) are less competent than the released evaluator (family decoded max dev BLEU-4 10.5 from joint-loss beam-3 and step-corrected, vs. 13.38). The remaining training runs either were not PHX-public-decoded (joint-loss supplementary, rescue, holdout families were not recoverable from disk after a host compromise and are omitted), produced degenerate output (α=1.0 distillation students, smallest ladder fractions), or were excluded for missing training logs. Without a competence-matched independent implementation, the causal checkpoint-identity question is unresolved. The human back-translations are a single-annotator resource without second-annotator adjudication; the 90.5% attenuation is conditional on one signer's interpretation and should not be taken as a population parameter (qualitative direction is stable across confidence tiers; SI Table S31). The probe response is a decoded-BT-text-metric effect, not a sign-language quality judgment; without target-conditioned DGS signer evaluation, semantic adequacy of the donor motion is unassessed.

6 Conclusion

This audit reports a failed reconstruction of the released SLRTP2025 BT evaluator from public artifacts under our disclosed implementation, and a descriptive non-reproduction of a constructed adversarial probe response.

For RQ1, none of our 14 near-faithful or 14 legacy-implementation reconstructions reached the released evaluator's competence; the source of the gap remains unidentified. For RQ2, all 61 of 61 non-degenerate constructible checkpoints give negative probe gaps (range $[-2.01, -0.21]$), but because every constructible checkpoint is less competent than the released evaluator (dev BLEU-4 6.5–10.5 vs. 13.38), we cannot distinguish whether this reflects the competence gap, an unrecovered training mechanism, or their interaction. For RQ3, reference replacement is associated with a gap reduction to $+0.95$ (interval includes zero) in a single-annotator, 461-item case study that cannot isolate reference-text defects from lexical-coupling disruption, paraphrase shift, or BLEU floor effects.

The scope is bounded: no causal checkpoint effect, no identified memorization mechanism, and no general leaderboard-rank instability is established. We frame the replay probe as an adversarial lexical-coupling stress test throughout. The audit's lasting contributions are the provenance, checkpoint-registry, and evaluator-audit infrastructure, transferable to future learned-evaluator audits.

Statements and Declarations

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Conflict of interest.

The authors declare no competing interests.

Data governance and ethics.

The datasets and model weights used in this audit are available under research-specific terms (not permissively public): PHOENIX-2014T is distributed under its own RWTH Aachen academic research licence; the SLRTP2025 pose bundle and released BT evaluator are governed by SLRTP2025 challenge terms; the Czehmann et al. Czehmann et al. (2026) back-translations are CC BY-NC-SA 4.0; CSL-Daily carries provider-specific academic terms. Pose tensors and decoded texts retain signer identity, biometric features, and signing style; we release per-item decoded text (low re-identification risk) and a curated subset of trained checkpoints needed for the

L1/L3 audit, but do *not* release per-signer raw poses. The artifact repository is a licence composition: root MIT Licence covers only the RCP audit software (scripts, source, Makefiles, paper sources); the Czehmann CSVs under `data/sacrebird/` are CC BY-NC-SA 4.0 (see `data/sacrebird/LICENSE` and `data/sacrebird/NOTICE`); the released BT evaluator under `checkpoints/released/` remains under SLRTP2025 challenge terms; other trained checkpoints inherit the upstream dataset restrictions. The CC BY-NC-SA ShareAlike clause may bind derivative redistributions of the Czehmann CSVs. We address four governance dimensions. (i) *Consent scope:* we conduct this research under the published licences; however, the original consent forms are not publicly available, so we cannot independently confirm that the informants' consent covers derivative replay-probe analysis. No new human-subjects data was collected. (ii) *Licensing:* the licence-composition table and per-directory attribution are documented in the artifact README. (iii) *Identity re-identification risk:* per-item decoded text and a curated subset of trained checkpoints are released; per-signer raw poses are not. (iv) *Deaf community involvement:* no Deaf/DGS stakeholders were involved in formulating the research questions or interpreting the results of this audit; we identify this as a limitation. The absence of Deaf co-investigators limits the ecological validity of our interpretations: the adversarial-probe framing, the reference-frame analysis, and the benchmark recommendations reflect a hearing-researcher perspective on a DGS resource, and Deaf-led sign-language AI research norms should guide any follow-up Desai et al. (2024). The Czehmann et al. back-translations we build upon were produced by a Deaf fluent L2 DGS signer.

Data availability.

All numerical values, per-checkpoint data, decoded hypotheses, and analysis scripts are in the artifact at https://anonymous.4open.science/r/rcp-audit-artifact-B314/README.md. The artifact is version-controlled (git commit `e523944`, 2026-08-11); a claim-to-file manifest mapping each headline number to its source file and verification command is at `claim_manifest.json` (artifact root), and a single-command consistency check (`make check-consistency`, invoking `scripts/check_paper_consistency.py`) verifies the paper's counts against the disk-scanned registry. The L1 audit environment is captured by `requirements-audit.txt` (4 packages: numpy, scipy, sacrebleu, pyyaml; install via `make install-audit`); the full `requirements.lock.txt` (236 packages) is needed only for L3 re-decoding and L4 from-scratch retraining. A persistent DOI-bearing Zenodo release will be deposited at submission, not deferred to acceptance. The supplementary material (`supplementary.pdf`) contains all SI appendices. Upstream datasets are available under research-specific terms documented in the artifact README and per-directory LICENSE/NOTICE files (PHOENIX-2014T under RWTH Aachen academic licence; Czehmann et al. human back-translations under CC BY-NC-SA 4.0, redistributed with attribution under those terms; CSL-Daily and SLRTP2025 under provider-specific academic terms).

Author contribution.

Both authors contributed to the study conception, design, data analysis, and manuscript preparation, and both read and approved the final manuscript. Specific contributions (CRediT taxonomy): **Xin Ji**: conceptualisation, methodology, software, investigation, formal analysis, data curation, writing—original draft, visualisation. **Cheng Zhang**: methodology, supervision, writing—review and editing, project administration.

Supplementary Information (SI)

The following supplementary materials, referenced throughout the main text, are available as a separate file (`supplementary.pdf`) and in the anonymous artifact. Each appendix is annotated with the question it answers.

- Sup. A: Missing Test ID Sensitivity — does the absent test item change the headline gap? (`sec:appendix_missing_id`)
- Sup. B: Per-seed Extension Cells 707–1405 — does the prediction interval hold beyond the six prospectively specified legacy seeds?
- Sup. C: Analysis Provenance — evidentiary role and multiplicity handling of each analysis. (`sec:provenance`)
- Sup. D: Canonical Checkpoint Registry — what was trained, and which families enter which claim? (`tab:checkpoint_registry`)
- Sup. E: Donor-Pool Substitution Decomposition — how much of the seen-pool advantage is pool-origin vs. similarity? (`sec:appendix_decomposition`, `tab:path_sensitivity`)
- Sup. F: Config-Faithful and Perturbation Details — protocol-mismatch check, weight-noise sweep, epoch-checkpoint decoupling search. (`sec:appendix_config_faithful`, `sec:appendix_perturbation`, `tab:perturbation`)
- Sup. G: Cluster-Aware Bootstrap — does the gap survive resampling by signer, broadcast-show, and date? (`sec:appendix_cluster_bootstrap`)
- Sup. H: Supplementary Oracle Diagnostics — is the evaluator-aware advantage robust to reference-overlap exclusion? (`sec:appendix_oracle`)
- Sup. I: Template-Density, Holdout Boundary, Donor-Exclusion — does the reversal persist on a clean template-holdout split and under donor-text exclusion? (`sec:appendix_template_boundary`)
- Sup. J: Transcript-Slot Diagnostic Details — do slot-level scores reproduce the BLEU reversal? (`sec:appendix_slot`)
- Sup. K: Competence Extrapolation and Per-Item Gap Decomposition — could competence in the unobserved interval explain the reversal, and which item features drive the gaps? (`sec:appendix_extrapolation`)
- Sup. L: CSL-Daily Boundary Control — does the readout pattern generalize beyond PHX-public? (`sec:csl-daily`)
- Sup. M: Leakage Sanity Checks — can the train-pool readout be explained by implementation shortcuts? (`sec:appendix_leakage`)
- Sup. N: Pre-Exclusion Materialization Sensitivity — how sensitive are the headline numbers to the donor-exclusion rule? (`sec:appendix_materialization`)
- Sup. O: BLEURT-Substitute Validation — does a learned metric corroborate the BLEU finding under reference replacement? (`sec:appendix_bleurt`)
- Sup. P: Main-Text Tables (SI Table S22–S33) — decoded-metric sensitivity, floor effect, cross-metric gaps, leaderboard, distillation ladder, checkpoint registry, protocol summary, rescue sweep, exposure estimands, reference sensitivity, and cross-evaluator re-decoding of noisy-gloss and USTC-MoE outputs. (`sec:appendix_moved_tables`)
- Sup. Q: Additional Protocol Details — tie-breaking, scorer equivalence, metric implementations, audit-design, reference-frame, and donor-origin figures (SI Fig. S4–S6). (`sec:appendix_moved_protocol`)
- Sup. R: Reconstruction Sanity Checks (SI Table S35) — small-sample overfit test, unified inference regression, and per-checkpoint teacher-forced vs. free-decode diagnostics ruling out several basic plumbing failures (data loading, masking, decoding alignment); does not rule out subtler training-objective or scheduler-timing defects (see main text §5). (`sec:appendix_sanity`)

Full checkpoint registry in SI Sup. D.

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Table 8: Checkpoint accounting, auto-generated from `results/accounting_table.json` (single source of truth, disk-scanned). "Runs" = from-scratch training executions; "Decoded" = beam-3 decoded on the canonical PHX-public gap panel; "Unique" = distinct SHA-256 weight binaries; "Non-deg." = non-degenerate (BLEU > 0, excluding 3 α=1.0 distillation students with empty output and 2 smallest ladder fractions). The near-faithful tier (14 corrected-recipe runs) provides recipe-faithfulness evidence; secondary (14 legacy, of which 6 prospectively specified) provides the prospectively specified non-reproduction direction plus stability; diagnostic is mechanism exploration.

Tier / Family	Runs	Decoded	Unique	Non-deg.
Near-faithful	14	14	14	14
Validation-freq-misread	4	4	4	4
Step-corrected (re-trained)	10	10	10	10
Secondary	14	14	14	14
Paper-derived (prospectively specified)	6	6	6	6
Paper-derived (extension seeds)	8	8	8	8
Diagnostic	40	38	37*	33
Joint-loss (greedy selection)	8	8	8	8
Joint-loss (beam-3 selection)	8	8	8	8
Joint-loss (supplementary, 1809/1810)	1	0	0	0
Confirmation	2	2	2	2
Rescue (dropout/wd variants)	1	0	0	0
Distillation	15	15	15	12
Ladder (data fractions)	4	4	4	2
Long-schedule	1	1	0*	1
Total trained	**68**	**66**	**65**	**61**

*One validation-freq-misread seed-202 artifact is byte-identical to a long-schedule artifact (same wandb run); it counts once in the unique-binary column (total unique = 66 decoded − 1 collision = 65). The released evaluator (1 checkpoint) is not a training run and is not included above. Eleven released-weight fine-tune variants (Gaussian noise, local perturbation) are not from-scratch training and are not counted here. The previously-reported large-arch, rescue-lr, cross-fit, and BT-retrained holdout families were not recoverable from disk after a host compromise (SI Sup. D) and are omitted; their family-level summary statistics had been retained in the legacy manual registry but cannot be reproduced from artifacts.

Table 9: Complete retrieval × construction design matrix explored on the released evaluator. Scores are corpus-level sacreBLEU-4 (0–100) on the canonical 641-item PHX-public set (REC=12.78). Of the 12 retrieval–construction cells, 6 were explored, 3 are not applicable ([a]), and 3 were not explored (*, †); full reasons in `results/probe_multiplicity.json`. The headline probe (bold) was selected as the maximal-gap explored cell; its CI is post-selection descriptive. Two random-donor baselines (below the midrule) lie outside the 4 × 3 matrix.

Retriever	Construction	Explored	Score	Gap vs. REC
Source-text Jaccard	**Pure donor copy**	✓	**23.02**	**+10.24**
Source-text Jaccard	PT-composed	✓	18.86	+6.08
Source-text Jaccard	Oracle-gloss-comp.	—	—	N/A[a]
LaBSE	Pure donor copy	✓	19.2	+6.42
LaBSE	PT-composed	—	—	not explored*
LaBSE	Oracle-gloss-comp.	—	—	N/A[a]
multi-SBERT	Pure donor copy	✓	16.5	+3.72
multi-SBERT	PT-composed	—	—	not explored*
multi-SBERT	Oracle-gloss-comp.	—	—	N/A[a]
Oracle gloss	Pure donor copy	✓	11.4	−1.38
Oracle gloss	PT-composed	✓	11.8	−0.98
Oracle gloss	Oracle-gloss-comp.	—	—	not explored†
Random-donor baselines (outside the 4 × 3 *matrix)*				
Random donor	Pure donor copy	✓	0.90	−11.88
Random donor	PT-composed	✓	0.77	−12.01

[a]Oracle-gloss-composed is definitionally tied to the oracle-gloss retriever, so it cannot be applied to a text-based retriever without changing the retriever. *Not explored; deemed low priority after Jaccard pure-copy dominated. †Subsumed by Oracle-gloss × PT-composed. Of the 12 retrieval–construction cells, 6 were explored, 3 are not applicable, and 3 were not explored. The headline probe was selected as the maximal-gap cell, so its CI is post-selection descriptive.

Table 10: Summary of the three primary descriptive comparisons. The released evaluator sits far outside t constructible family range on all three views. Family n: (i) 66 beam-3-decoded training runs (range over 61 no degenerate; 5 degenerate excluded); (ii) 30 constructible checkpoints with uniform full-pool readout (released liste separately); (iii) gate-eligible non-distillation training runs. Per-view family composition in SI Sup. D.

Comparison	Family n	Released evaluator	Family range	Released − family ma
(i) Gap–competence	66	+10.24 at dev 13.38	[−2.01, −0.21] (61 non-deg.)	+10.4
(ii) Train-pool readout BLEU	30	78.8 (70.7% EM)	6.6–9.3 (1–2% EM)	+69
(iii) Competence gate	52	13.38	6.5–10.5 (uniform; joint-loss beam-3 max 10.5)	+2.8

Table 11: Canonical fixed-output evaluation on the released 641-pose PHX-public subset (released evaluator and six prospectively specified legacy seeds (101–606); eight extension seeds in SI Table S27). PURE is TN-PURE-v1; COMP is TN-PTCOMP-v1. Gaps are relative to local REC.

Evaluator	REC	PURE	Gap	COMP	Gap
Released	12.78	23.02	**+10.24**	18.86	+6.09
Seed 101	9.24	8.45	−0.80	7.92	−0.65
Seed 202	8.22	6.92	−1.30	7.54	−1.28
Seed 303	9.10	7.25	−1.85	8.71	−0.96
Seed 404	9.52	7.71	−1.81	6.38	−0.81
Seed 505	9.71	8.44	−1.28	7.97	−1.08
Seed 606	7.55	6.31	−1.24	7.93	−0.16
Extension 707–1405 (range)	see SI Table S27		−1.48 to −0.87		−1.03 to −0.44